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概率答案

1.5 (a) $\bar{E}F\bar{G}$

(b) $EF\bar{G}$

(c) $\overline{\bar{E}\bar{F}\bar{G}} = E\cup F\cup G$

或 $E\bar{F}\bar{G}\cup\bar{E}F\bar{G}\cup\bar{E}\bar{F}G\cup EF\bar{G}\cup E\bar{F}G\cup\bar{E}FG\cup EFG$

(d) $EF\bar{G}\cup E\bar{F}G\cup\bar{E}FG\cup EFG$ 或 $EF\cup EG\cup FG$ (至少发生2个)

(e) EFG

(f) $\bar{E}\bar{F}\bar{G} = \overline{E\cup F\cup G}$

为对立事件

(g) $E\bar{F}\bar{G}\cup\bar{E}F\bar{G}\cup\bar{E}\bar{F}G\cup\bar{E}\bar{F}\bar{G}$ 或 $\overline{EF\cup EG\cup FG}$ (至多发生1个), 所以

(h) $\overline{E\cap F\cap G}$ 或 $\bar{E}\bar{F}\cup\bar{E}\bar{G}\cup\bar{F}\bar{G}$ 注: 有的同学区分不开

或 $EF\bar{G}\cup E\bar{F}G\cup\bar{E}FG\cup E\bar{F}\bar{G}\cup\bar{E}F\bar{G}\cup\bar{E}\bar{F}G\cup\bar{E}\bar{F}\bar{G}$

或 $(E\cup F\cup G)-(E\cap F\cap G) = (E\cup F\cup G)\cap\overline{EFG}$

$\overline{E\cup F} = \bar{E}\cup\bar{F}$

与 $\overline{\bar{E}\bar{F}} = E\cap F$

日有同学用"+"表示"$\cup$"

日有的同学用 $\bigcirc$ EFG → Ω

1.6 (a) $(A\cup B)\cap(B\cup C) = B\cup(A\cap C)$

(b) $(A\cup B)\cap(A\cup\bar{B}) = A\cup(B\cap\bar{B}) = A\cup\phi = A$

(c) $\underbrace{(A\cup B)\cap(A\cup\bar{B})}_{为A}\cap(\bar{A}\cup B) = A\cap(\bar{A}\cup B) = A\cap B$

1.7 No.

They are not disjoint, because the event A and B can occur at the same time, which means you can obtain a head on the first toss, and you also obtain a head on the sixth toss.

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1.11 (1) In case a

$\#\Omega = 2^3 = 8$

$B_1 = \{(B,G,G), (G,B,G), (G,G,B)\}$

$C = \{(B,G,G), (G,B,G), (G,G,B), (G,G,G)\}$

$\therefore P(B_1) = \frac{\#B_1}{\#\Omega} = \frac{3}{8}$

$P(C) = \frac{\#C}{\#\Omega} = \frac{4}{8} = \frac{1}{2}$

(2) In case b

$\Omega = \{\overset{\frac{1}{2}}{(G)}, \overset{\frac{1}{2}\times\frac{1}{2}}{(B,G)}, \overset{\frac{1}{8}}{(B,B,G)}, \overset{\frac{1}{8}}{(B,B,B)}\}$

$\therefore B_1 = \{(B,G)\}$

$C = \{(G)\}$

$\therefore P(B_1) = \frac{1}{4}$, $P(C) = \frac{1}{4}$ (?) $\frac{1}{2}$

(3) In case C

$\Omega = \{\overset{\frac{1}{4}}{(B,G)}, \overset{\frac{1}{4}}{(G,B)}, \overset{\frac{1}{8}}{(B,B,G)}, \overset{\frac{1}{8}}{(G,G,B)}, \overset{\frac{1}{8}}{(B,B,B)}, (G$

? $P(B_1) = \frac{3}{6} = \frac{1}{2}$

? $P(C) = \frac{2}{6} = \frac{1}{3}$

$P(B_1) = \frac{1}{4} + \frac{1}{4} + \frac{1}{8}$

$P(C) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$

13. Solution: Let y and z be the time ~~the~~ a train ~~and~~ ~~the~~ a bus arriving the station, respectively, then our sample space is a square $\Omega = \{(y,z) \mid 0 \le y, z \le 60\}$. Let M be the event that ~~the~~ a train and a bus will meet at the station, then M will occur if and only if $-x \le z - y \le 10$,

$$M = \{(y,z) \mid 0 \le y, z \le 60,\ -x \le z - y \le 10\}$$

M is shown as the shaded in the following ~~picture~~ figure.

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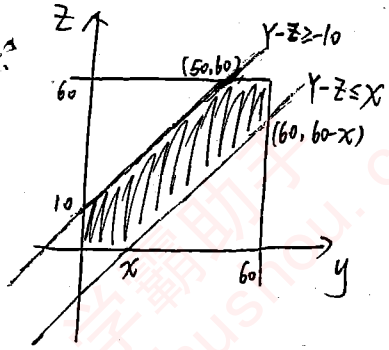
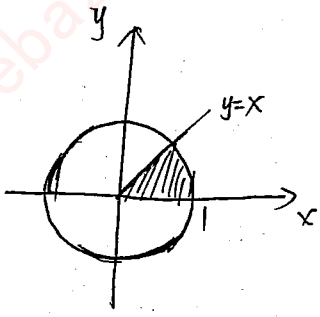


$$p(M)=1-\frac{\frac{1}{2}\times 50^2+\frac{1}{2}(60-x)^2}{60^2}=0.5$$

$$\Rightarrow \quad x=60-10\sqrt{11}$$
$$\approx 26.834$$

15.



Let M be the event that a picked point lies in the angular sector from 0 to $\pi/4$ radians.

M is shown as the shaded in the above figure.

So $\quad p(M)=\frac{1}{8}$

注：所的图形形状为单位圆周上任取一点，该点落在0到$\frac{\pi}{4}$圆弧上的概率为 $p=\frac{\frac{\pi}{4}}{2\pi}=\frac{1}{8}$

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22.

$P(AB)=P(A)+P(B)-P(A\cup B)$

$P(B)<P(A)$

① 当 $B\subset A$ 时，则 $P(A\cup B)=P(A)=\frac{3}{4}$ $\therefore P(AB)=\frac{1}{3}$

② 当 $A\cup B=\Omega$ 时，则 $P(A\cup B)=1$ $\therefore P(AB)=\frac{1}{12}$

$\therefore$ $P(AB)$ 的两个极限值都有可能取到

$P(A\cup B)=P(A)+P(B)-P(AB)$

$=\frac{3}{4}+\frac{1}{3}-P(AB)$ 而 $P(AB)\in[\frac{1}{12},\frac{1}{3}]$

$\therefore P(A\cup B)\in[\frac{3}{4},1]$

25.

$\because P(A)=P(AB)+P(A\bar{B})$

$\therefore P(AB)=P(A)-P(A\bar{B})=0.5-0.32=0.18$

$P(A\cup B)=P(A)+P(B)-P(AB)=0.5+0.4-0.18=0.72$

26.

$P(AB)=P(A)+P(B)-P(A\cup B)=p+q-r$

$P(A\bar{B})=P(A)-P(AB)=p-(p+q-r)=r-q$

$P(\bar{A}\bar{B})=P(\bar{B})-P(A\bar{B})=(1-P(B))-P(A\bar{B})=(1-q)-(r-q)=1-r$

28.

$\because AB=\phi$ $\therefore P(AB)=0$

$\because P(A|\bar{B})+P(\bar{A}|\bar{B})=1$

$P(A|\bar{B})=P(\bar{A}|\bar{B})$

$\therefore P(A|\bar{B})=P(\bar{A}|\bar{B})=\frac{1}{2}$

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$$P(A|\bar{B}) = \frac{P(A\bar{B})}{P(\bar{B})} = \frac{P(A)-P(AB)}{1-P(B)} = \frac{a}{1-a} = \frac{1}{2}$$

$\therefore a = \frac{1}{3}$

$$P(A\cup B) = P(A)+P(B)-P(AB) \xlongequal{P(AB)=0} P(A)+P(B) = 2a = \frac{2}{3}$$

注：有很多同学把 $AB=\phi$，理解成 $P(AB)=P(A)\cdot P(B)$

进而推出 $a=\frac{1}{2}$

31. (此题做第2问的器的相对的些)

(a) ~~Denote the event~~ "

Let A denote the event that "The second ball drawn is ~~green~~"

$$\therefore P(A) = \underbrace{\frac{b}{b+c}\cdot\frac{c}{b+c+d}}_{\text{第一个被取到的球是绿色的}} + \underbrace{\frac{c}{b+c}\cdot\frac{c+d}{b+c+d}}_{\text{第一个被取到的球是蓝绿色的.}} = \frac{c}{b+c}$$

(b) Let B denote the event that "The first ball drawn is blue"

$$\therefore P(B|A) = \frac{P(AB)}{P(A)} = \frac{\frac{c}{b+c}\cdot\frac{c+d}{b+c+d}}{\frac{c}{b+c}} = \frac{c+d}{b+c+d}$$

33

一. (a) Let A denote the event that "both bulbs are defective".

题意是这个解答

$$\therefore P(A) = \underbrace{\frac{1}{2}\cdot\frac{C_{100}^2}{C_{1000}^2}}_{\text{两边品来自第一个箱子}} + \underbrace{\frac{1}{2}\cdot\frac{C_{100}^2}{C_{2000}^2}}_{\text{两边品来自第二个箱子.}} = \frac{1}{2}\times\frac{100}{1000}\times\frac{99}{999} + \frac{1}{2}\times\frac{100}{2000}\times\frac{99}{1999} = 0.00619$$

(b) Let B denote the event that "two defective bulbs came from box 1"

$$\therefore P(B|A) = \frac{P(AB)}{P(A)} = \frac{\frac{1}{2}\cdot\frac{C_{100}^2}{C_{1000}^2}}{P(A)} = 0.8$$

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二. 注：还有不少同学给的解法：先从2个Box中任选一个，从中抽到3次品；再从2个Box中任选一个，又从中抽到3次品。

得到：

(a) $$P(A)=\underbrace{\frac{100}{1000}\times\frac{99}{999}}_{\text{2个次品均来自Box1}}+\underbrace{\frac{100}{2000}\times\frac{99}{1999}}_{\text{2个次品均来自Box2}}+\underbrace{\frac{100}{1000}\times\frac{100}{2000}}_{\text{一个来自Box1，另一个来自Box2}}\approx 0.017$$

(b) $$P(B|A)=\frac{\frac{100}{1000}\times\frac{99}{999}}{P(A)}\approx 0.57$$

38.

(a) $$P(A)=\frac{6}{10}\times\frac{8}{11}=\frac{24}{55}$$

(b) $$P(B)=\frac{4}{10}\times\frac{4}{11}=\frac{8}{55}$$ （B表示从第一个Box中选一个绿球的第2个，然后从第二个Box选了一个绿球的同第一个色）

$$\therefore P(C)=P(A)+P(B)=\frac{32}{55}$$ （C表示两个箱子中最后红球与绿球个数最初的情况相同）.

注：此题做错的人很少.

39.

A: one person has the disease.

B: one person test positive.

(a) $$P(B)=P(A)P(B|A)+P(\bar{A})P(B|\bar{A})$$

$$=0.5\times(1-2\%)+(1-0.5)\times 3\%=\frac{101}{200}$$

(b) $$P(A|B)=\frac{P(AB)}{P(B)}=\frac{\frac{1}{2}\times(1-2\%)}{\frac{101}{200}}=\frac{98}{101}$$

注：one half percent：被同学给分成了50%，0.5%，1.5%，上面答案是50%情况，下面两种就是将上面的50%替换了下.

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1.46. ∵A与B独立 ∴ $P(\bar{A}\bar{B})=P(\bar{A})P(\bar{B})=[1-P(A)]P(\bar{B}) \xlongequal{\because P(A)=P(B)} [1-P(\bar{B})]P(\bar{B})=\frac{1}{9}$

∴ $P^2(\bar{B})-P(\bar{B})+\frac{1}{9}=0$

∴ $P(A)=P(\bar{B})=\frac{3\pm\sqrt{5}}{6}$

1.47. 用X表示n次独立重复试验中事件A发生的次数，且P(A)=p，(则 $X\sim B(n,p)$)
此记号后面学，可不写

(a) $P(X\geq 2)=1-P(X=0)-P(X=1)$

$=1-C_n^0p^0(1-p)^n-C_n^1p(1-p)^{n-1}$

$=1-(1-p)^n-np(1-p)^{n-1}$

$=1-(1-p)^{n-1}[1+p(n-1)]$

(b) $P(X\geq 3)=P(X\geq 2)-P(X=2)$ $\left(\xlongequal{或} 1-P(X=0)-P(X=1)-P(X=2)\right)$

$=1-(1-p)^n-np(1-p)^{n-1}-\frac{n(n-1)}{2}p^2(1-p)^{n-2}$

50. 设X表示男孩个数，则($X\sim B(3,p)$)，其中p表示生男孩的概率，则
此记号后面学，可不写

$P(A)=P(X=0)+P(X=3)=C_3^0p^0(1-p)^3+C_3^3p^3(1-p)^0=(1-p)^3+p^3$

$P(B)=P(X=0)+P(X=1)=C_3^0p^0(1-p)^3+C_3^1p(1-p)^2=(1-p)^3+3p(1-p)^2$

$P(C)=1-P(A)=1-(1-p)^3-p^3$

$P(AB)=P(X=0)=(1-p)^3$

∵ $AC=\phi$ ∴ $P(AC)=0$

$P(BC)=P(X=1)=C_3^1p(1-p)^2=3p(1-p)^2$

(a) $p=\frac{1}{2}$时，

$P(A)=\frac{1}{4}$，$P(B)=\frac{1}{2}$，$P(C)=\frac{3}{4}$，$P(AB)=\frac{1}{8}$，~~$P(AC)=0$~~，$P(BC)=\frac{3}{8}$

∴ $P(AB)=P(A)P(B)$，$P(BC)=P(B)P(C)$

∴ A与B独立，B与C独立.

(b) ∵ $P(AC)=0$，$P(A)P(C)=\frac{1}{4}\cdot\frac{3}{4}=\frac{3}{16}$ ∴ $P(AC)\neq P(A)P(C)$ ∴ A与C不独立.

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(c) 当 $p=\frac{1}{2}$ 时，则 $P(AB)\neq P(A)\cdot P(B)$，

$P(BC)\neq P(B)\cdot P(C)$

且仍有 $P(AC)\neq P(A)\cdot P(C)$

∴ 结论(a)不成立，结论(b)仍成立.

(d) 此时 $X\sim B(4,\frac{1}{2})$ （此纪号后面学，可不写）

$P(A)=P(X=0)+P(X=4)=\frac{2}{2^4}=\frac{1}{8}$

$P(B)=P(X=0)+P(X=1)=C_4^0(\frac{1}{2})^0(\frac{1}{2})^4+C_4^1\frac{1}{2}(\frac{1}{2})^3=\frac{5}{16}$

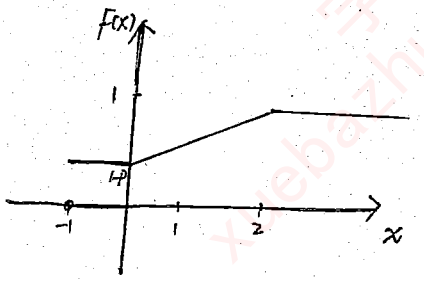
$P(C)=1-P(A)=\frac{7}{8}$

$P(AB)=P(X=0)=\frac{1}{16}$，$P(AC)=0$，$P(BC)=P(X=1)=C_4^1\frac{1}{2}(\frac{1}{2})^3=\frac{4}{16}=\frac{1}{4}$

∴ $P(AB)\neq P(A)P(B)$，$P(AC)\neq P(A)P(C)$，$P(BC)\neq P(B)P(C)$

∴ 结论(a)不成立，结论(b)仍成立.

2.2.



(a) $P(X=-1)=F(-1)-F(-1^-)=1-p$

(b) $P(X=0)=F(0)-F(0^-)=(1-p)-(1-p)=0$ （不少同学得到 $P(X=0)=1-p$ ）

(c) $P(X>1)=1-P(X\leq 1)$

$=1-(1-p+\frac{p}{2})=\frac{p}{2}$

2.4. (a) $P(\frac{1}{2}\leq X\leq\frac{3}{2})=P(X\leq\frac{3}{2})-P(X<\frac{1}{2})=F(\frac{3}{2})-F(\frac{1}{2}^-)=\frac{1}{2}\cdot\frac{3}{2}-\frac{1}{3}\cdot\frac{1}{2}=\frac{3}{4}-\frac{1}{6}=\frac{7}{12}$

(b) $P(\frac{1}{2}\leq X\leq 1)=P(X\leq 1)-P(X<\frac{1}{2})=F(1)-F(\frac{1}{2}^-)=\frac{1}{2}-\frac{1}{3}\cdot\frac{1}{2}=\frac{1}{2}-\frac{1}{6}=\frac{1}{3}$

(c) $P(\frac{1}{2}\leq X<1)=P(X<1)-P(X<\frac{1}{2})=F(1^-)-F(\frac{1}{2}^-)=\frac{1}{3}-\frac{1}{3}\cdot\frac{1}{2}=\frac{1}{6}$

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(d) $P(1\le X\le \frac{3}{2}) = P(X\le \frac{3}{2}) - P(X<1) = F(\frac{3}{2}) - F(1^-) = \frac{1}{2}\cdot\frac{3}{2} - \frac{1}{3} = \frac{5}{12}$

(e) $P(1<X<2) = P(X<2) - P(X\le 1) = F(2^-) - F(1) = 1-\frac{1}{2} = \frac{1}{2}$

2.6. 当掷硬币次数 $X=n$ 时，那么在前 $n-1$ 次掷硬币实验中出现正面朝上的次数有 $r-1$ 次，反面朝上的是 $(n-1)-(r-1)=n-r$ 次.

$\therefore P(X=n) = C_{n-1}^{r-1} p^{r-1}(1-p)^{n-r}\cdot p = C_{n-1}^{r-1} p^{r}(1-p)^{n-r}$.

2.8.

$\frac{a}{8}+\frac{a}{4}+\frac{a}{8} = 1 \quad \therefore a=2$

2.10.

$P(X=0) = F(0)-F(0^-) = \frac{1}{2}$

$P(X=1) = F(1)-F(1^-) = \frac{3}{5}-\frac{1}{2} = \frac{1}{10}$

$P(X=2) = F(2)-F(2^-) = \frac{4}{5}-\frac{3}{5} = \frac{1}{5}$

$P(X=3) = F(3)-F(3^-) = \frac{9}{10}-\frac{4}{5} = \frac{1}{10}$

$P(X\ge 3.5) = 1-P(X<3.5) = 1-F(3.5^-) = 1-\frac{9}{10} = \frac{1}{10}$

2.12 该题假设不对.

(a) 若 $f(x)$ 为密度函数，则 $\int_{-\infty}^{+\infty} f(x)dx = 1$.

当 $d>1$ 时，$\int_1^{+\infty} cx^{-d}dx = \frac{c}{1-d}x^{1-d}\Big|_1^{+\infty} = \frac{c}{d-1} = 1 \quad \therefore c=d-1$ 且此时 $f(x)$ 为概率密度函数

则 $x>1$ 时，$F(x) = \int_{-\infty}^{x} f(y)dy = \int_{-\infty}^{x}(d-1)x^{-d}dx = 1-x^{1-d}$

$$\therefore F(x) = \begin{cases} 1-x^{1-d} & x>1 \\ 0 & \text{otherwise} \end{cases}$$

当 $d\le 1$ 时，$f(x)$ 不是概率密度函数.

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(b) $\int_{-\infty}^{+\infty} f(x)dx = \int_{-\infty}^{+\infty} c\frac{e^x}{(1+e^x)^2}dx = -\frac{c}{1+e^x}\Big|_{-\infty}^{+\infty} = -c[0-1] = c = 1$

$f(x)$ 为概率密度函数，且 $c=1$

$\therefore F(x) = \int_{-\infty}^{x} f(y)dy = \int_{-\infty}^{x} \frac{e^y}{(1+e^y)^2}dy = -\frac{1}{1+e^y}\Big|_{-\infty}^{x} = 1 - \frac{1}{1+e^x}$

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16. 解:

X	-1	0	1
P	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$

$E(X)=-1\times\frac{1}{3}+0\times\frac{1}{3}+1\times\frac{1}{3}=0$

$Var(X)=EX^2-(EX)^2$

$=EX^2$

$=(-1)^2\times\frac{1}{3}+1^2\times\frac{1}{3}=\frac{2}{3}$

$SD(X)=\sqrt{VarX}=\frac{\sqrt{6}}{3}$

25 解: 由 $\int_{-\infty}^{+\infty}f(x)dx=\int_0^1(ax+bx^2)dx=1$

$EX=\int_{-\infty}^{+\infty}xf(x)dx=\int_0^1x(ax+bx^2)dx=0.6$

得 $a=3.6$, $b=-2.4$

(a) $P(X<\frac{1}{2})=\int_0^{\frac{1}{2}}(3.6x-2.4x^2)dx=0.35$

(b) $Var(X)=E(X^2)-(EX)^2=\int_0^1x^2(3.6x-2.4x^2)dx-0.36=0.06$

28 解: (a) $P(X>2)=1-P(X\le2)=1-F(2)=1-(1-e^{-4})=e^{-4}$

(b) $P(1<X<3)=P(X<3)-P(X\le1)=F(3)-F(1)=(1-e^{-9})-(1-e^{-1})=e^{-1}-e^{-9}$

(c) $EX=\int_0^{+\infty}xdF(x)=\int_0^{+\infty}xd(1-e^{-x^2})=\int_0^{+\infty}xd(e^{-x^2})=-xe^{-x^2}\Big|_0^{+\infty}+\int_0^{+\infty}e^{-x^2}dx$

$=\frac{\sqrt{\pi}}{2}$

(d) $EX^2=\int_0^{+\infty}x^2dF(x)=-\int_0^{+\infty}x^2de^{-x^2}=-x^2e^{-x^2}\Big|_0^{+\infty}+2\int_0^{+\infty}xe^{-x^2}dx$

$=-x^2e^{-x^2}\Big|_0^{+\infty}-e^{-x^2}\Big|_0^{+\infty}=1$

注: 不少同学的(c)与(d)都没得到最后结果, 不会积分.

$\therefore Var(X)=EX^2-(EX)^2=1-\frac{\pi}{4}$

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18 解：$\sum_{k=1}^{+\infty}|x_k|P(x_k)=\sum_{k=1}^{+\infty}2^{2k}\cdot\frac{1}{2^{2k}}+\sum_{k=1}^{+\infty}|-2^{2k-1}|\cdot\frac{1}{2^{2k-1}}=\sum_{k=1}^{+\infty}1+\sum_{k=1}^{+\infty}1=\infty$

$\therefore$ EX不存在.

注：有的同学写：$\sum_{k=1}^{+\infty}x_kP(x_k)=\sum_{k=1}^{+\infty}2^{2k}\frac{1}{2^k}+\sum_{k=1}^{+\infty}(-2^{2k})\cdot$

$=\infty-\infty$

$\therefore$ EX不存在.

19 解：(a) $\int_{-\infty}^{+\infty}f(x)dx=\int_0^1(a+bx^2)dx=a+\frac{b}{3}=1$

$EX=\int_0^1x(a+bx^2)dx=\frac{a}{2}+\frac{b}{4}=0.6$

解得：$a=0.6$，$b=1.2$

(b) $f(x)=\begin{cases}0.6+1.2x^2 & 0\le x\le 1\\ 0 & \text{otherwise}\end{cases}$

$F_Y(y)=P(Y\le y)=P(5X+2\le y)=P(X\le\frac{y-2}{5})=\int_{-\infty}^{\frac{y-2}{5}}f(x)dx$

当 $\frac{y-2}{5}<0$，即 $y<2$ 时，$F_Y(y)=0$

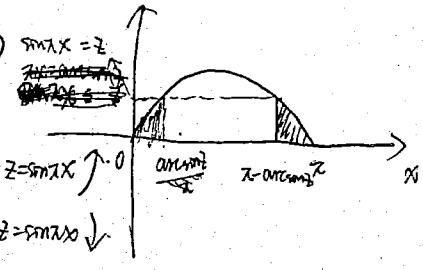
当 $0\le\frac{y-2}{5}\le1$，即 $2\le y\le7$ 时，$F_Y(y)=\int_0^{\frac{y-2}{5}}(0.6+1.2x^2)dx=\frac{2(y-2)^3+75(y-2)}{625}$

当 $\frac{y-2}{5}>1$，即 $y>7$ 时，$F_Y(y)=1$

$\therefore f_Y(y)=\begin{cases}\frac{6(y-2)^2+75}{625}=0.12+0.0096(y-2)^2 & 2\le y\le7\\ 0 & \text{否则}\end{cases}$

注：很多同学没讨论 y 取值情况，直接写成了 $2\le y\le7$ 时的那一种

(c) $\sin\pi x=z$



当 $0\le x\le\frac{1}{2}$ 时，$z=\sin\pi x$ ↗

当 $\frac{1}{2}<x\le1$ 时，$z=\sin\pi x$ ↘

$\{\sin\pi x\le z\}=\{x\le\frac{\arcsin z}{\pi}\}\cup\{x\ge1-\frac{\arcsin z}{\pi}\}$

$\therefore$ 当 $x\in[0,1]$ 时，有 $z=\sin\pi x\in[0,1]$

有 $F_Z(z)=P(\sin\pi x\le z)=P(x\le\frac{\arcsin z}{\pi})+P(x\ge1-\frac{\arcsin z}{\pi})$

$=\int_0^{\frac{\arcsin z}{\pi}}(0.6+1.2x^2)dx+\int_{1-\frac{\arcsin z}{\pi}}^{1}(0.6+1.2x^2)dx$

$\therefore$ 此时，$f_Z(z)=[0.6+1.2(\frac{\arcsin z}{\pi})^2]\frac{1}{\pi}\frac{1}{\sqrt{1-z^2}}+[0.6+1.2(1-\frac{\arcsin z}{\pi})^2]\frac{1}{\pi}\cdot\frac{1}{\sqrt{1-z^2}}$

$=\frac{1.2}{\pi}\frac{1}{\sqrt{1-z^2}}[1+(\frac{\arcsin z}{\pi})^2+(1-\frac{\arcsin z}{\pi})^2]$

注：以多所有同学都只考虑了 $\{x\le\frac{\arcsin z}{\pi}\}$ 这一种情况，也就是说他们把 $Z=\sin\pi X$ 按照、学习过的处理了.

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$$\therefore f_Z(z)=\begin{cases}\frac{1.2}{\pi}\frac{1}{\sqrt{1-z^2}}\left[1+\left(\frac{\arcsin z}{\pi}\right)^2+\left(1-\frac{\arcsin z}{\pi}\right)^2\right] & 0\le z\le 1\\ 0 & \text{otherwise}\end{cases}$$

(d)

$\cancel{E(Y)=\int_2^7 \frac{6(y-2)^2+75}{625}dy}$

$=5E(X)+2=5\cdot 0.6+2=5$

$E(Y)=E(5X+2)=\int_0^1(5x+2)f_X(x)dx=\int_0^1(5x+2)(0.6+1.2x^2)dx=5$

$E(Z)=E(\sin\pi X)=\int_0^1\sin\pi x\cdot(0.6+1.2x^2)dx=\frac{2.4}{\pi}-\frac{4.8}{\pi^3}$

21 解：当 $y\ge 1$ 时，$F_Y(y)=P(Y\le y)=P(|X|+1\le y)=P(|X|\le y-1)$

$=P(-y+1\le X\le y-1)$

$=F_X(y-1)-F_X(-y+1)$

$\therefore f_Y(y)=f_X(y-1)+f_X(-y+1)$

注：有不少同学这样做的.

当 $X\ge 0$ 时，$Y=X+1$，则 $f_Y(y)=f_X(y-1)$

当 $X<0$ 时，$Y=-X+1$，则 $f_Y(y)=f_X(-y+1)$

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31 (a) $X \sim B(17, 0.05)$ $\lambda = np = 0.85$

$P(X=1) = C_{17}^{1}\, 0.05 \cdot (0.95)^{16} \approx \frac{0.85}{1!} e^{-0.85} = 0.85 e^{-0.85} \approx 0.363$

(b) $P(X \le 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$

$= e^{-0.85}\left(\frac{0.85^0}{0!} + \frac{0.85}{1!} + \frac{0.85^2}{2!} + \frac{0.85^3}{3!}\right) \approx 0.989$

(c) $P(X \ge 2) = 1 - P(X=0) - P(X=1)$

$= 1 - e^{-0.85}\left(\frac{0.85^0}{0!} + \frac{0.85}{1!}\right) \approx 0.209$

~~注:~~

注: 36、37 的题目应当用泊松极限定理

40 $X \sim B(12, 0.1)$, $\lambda = 1.2$

(1) $P(X \le 2) = P(X=0) + P(X=1) + P(X=2)$

$= C_{12}^{0}(0.9)^{12} + C_{12}^{1}\, 0.1 \cdot (0.9)^{11} + C_{12}^{2}(0.1)^2 \cdot (0.9)^{10}$

$= e^{-1.2}\left(1 + 1.2 + \frac{1.2^2}{2!}\right) \approx 0.88$

(2) $P(X \ge 3) = 1 - P(X \le 2) = 0.12$

$Y \sim B(6, 0.12)$

$\therefore P(Y \ge 1) = 1 - P(Y=0) = 1 - C_6^0 (0.12)^0 \cdot (0.88)^6 \approx 0.54$

(c) $X \sim B(6, 0.12)$

$E(X) = np = 0.72$, $var(X) = np(1-p) = 6 \times 0.12 \times 0.88 = 0.6336 = \sigma^2$

$\Rightarrow \sigma \approx 0.796$

$P(X - 0.72 \ge 2 \times 0.796)$

$= P(X \ge 2.3) = P(X \ge 3) = 1 - P(X \le 2) \approx 1 - \left[C_6^0 (0.12)^0 (0.88)^6 + C_6^1\, 0.12 \cdot (0.88)^5 + C_6^2 (0.12)^2 \cdot (0\right.$

≈ 0.026

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42 $X \sim Ge(p)$, $P_X(x) = (1-p)^{x-1} \cdot p$

(a) 令 $Y = X^2$

$P_Y(y) = P(Y=y) = P(X=\sqrt{y}) = (1-p)^{\sqrt{y}-1} \cdot p$

(b) 令 $Y = X+3$

$P_Y(y) = P(X+3=y) = P(X=y-3) = (1-p)^{y-4} \cdot p$

44 解: $X \sim B(50, 0.05)$, $\lambda = np = 2.5$

$\therefore P(X \le 2) = P(X=0) + P(X=1) + P(X=2)$

$= C_{50}^0 (0.95)^{50} + C_{50}^1 0.05 \cdot (0.95)^{49} + C_{50}^2 (0.05)^2 \cdot (0.95)^{48}$

$\approx e^{-2.5} \left(\frac{2.5^0}{0!} + \frac{2.5}{1!} + \frac{2.5^2}{2!} \right) \approx 0.5438$

46 解: Let X be the number of winning the prize.

$X \sim B(50, \frac{1}{100})$ $\lambda = np = 0.5$

(a) $P(X \ge 1) = 1 - P(X=0) = 1 - C_{50}^0 (0.99)^{50} = 1 - e^{-0.5} \cdot \frac{0.5^0}{0!} \approx 0.3935$

(b) $P(X=1) = C_{50}^1 0.01 \cdot (0.99)^{49} \approx e^{-0.5} \cdot 0.5 \approx 0.3033$

(c) $P(X \ge 2) = 1 - P(X=0) - P(X=1) \approx 0.0902$

$= P(X \ge 1) - P(X=1)$

47. 解: $X \sim U(-1, 1)$, 则 $F_X(x) = \begin{cases} \frac{x+1}{2} & -1 < x < 1 \\ 1 & x \ge 1 \\ 0 & x \le -1 \end{cases}$

$Y = X^2$

当 $y \le 0$ 时, $F_Y(y) = 0$

当 $y > 0$ 时, $F_Y(y) = P(X^2 \le y) = P(-\sqrt{y} \le X \le \sqrt{y}) = F(\sqrt{y}) - F(-\sqrt{y}) = \begin{cases} \sqrt{y} & 0 < y < 1 \\ 1 & y \ge 1 \end{cases}$

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$\therefore F_Y(y)=\begin{cases}\sqrt{y} & 0<y<1\\ 1 & y\geq 1\\ 0 & y\leq 0\end{cases}$

2.47 解：$X\sim U(0,1)$，则 $f_X(x)=\begin{cases}1, & 0<x<1\\ 0 & 其它.\end{cases}$

$Y=-\ln X$，$x\in(0,1)$时，$y\in(0,+\infty)$，且y关于x单调，则$x=e^{-y}$

$\therefore f_Y(y)=\begin{cases}f_X(e^{-y})\,|-e^{-y}| \\ 0\end{cases}=\begin{cases}e^{-y}, & y>0\\ 0, & y\leq 0\end{cases}$

注：46，50中在标准的取值范围处，本题除了的是x的取范围.

50. 解：$X\sim exp(\lambda)$，则 $f_X(x)=\begin{cases}\lambda e^{-\lambda x}, & x\geq 0\\ 0, & x<0\end{cases}$，$F_X(x)=\begin{cases}1-e^{-\lambda x}, & x\geq 0\\ 0, & x<0\end{cases}$

(1) $Y=e^X$，当$x\geq 0$时，$y\geq 1$，且y关于x单调，则$x=\ln y$

$\therefore f_Y(y)=\begin{cases}f_X(\ln y)\cdot(\frac{1}{y}) \\ 0\end{cases}=\begin{cases}\lambda e^{-\lambda \ln y}\cdot\frac{1}{y} \\ 0\end{cases}=\begin{cases}\lambda y^{-\lambda-1}, & y\geq 1\\ 0, & y<1\end{cases}$

(2) $Y=\min\{X,3\}$

$y\geq 3$ $P(Y\leq y)=P\{\min\{X,3\}\leq y\}$

$=1-P\{\min\{X,3\}>y\}$

$=1-P(X>y,3>y)$

$=1$

$F_Y(y)=\begin{cases}1-e^{-\lambda y}, & 0\leq y<3\\ \cancel{1-e^{-3\lambda}}\ 1, & y\geq 3\\ 0, & y<0\end{cases}$

看作你有时间做下(2)吧，不考虑没做.

$\therefore f_Y(y)=\begin{cases}\lambda e^{-\lambda y}, & 0\leq y<3\\ 0 & 其它\end{cases}$

$0\leq y<3$，$P(Y\leq y)=1-P(X>y,3>y)$

$=1-P(X>y)=1-e^{-\lambda y}$

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2.53 解: $X\sim N(1,2^2)$ 54

(a) $P(X\le 3)=P(\frac{X-1}{2}\le\frac{3-1}{2})=P(\frac{X-1}{2}\le 1)=\Phi(1)$

(b) $P(X>1.5)=1-P(X\le 1.5)=1-P(\frac{X-1}{2}\le\frac{1.5-1}{2})=1-\Phi(0.25)$

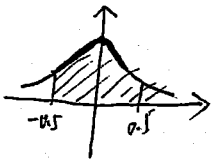
(c) $P(X=1)=0$ 注: 不少同学写了这种答案: 错误 $\begin{cases}①\ P(X=1)=P(Z=0)=\Phi(0)=\frac{1}{2}\\ ②\ P(X=1)\approx f_X(1)=\frac{1}{2\sqrt{2\pi}}e^{-\frac{(1-1)^2}{2\times 4}}=\frac{1}{2\sqrt{2\pi}}\end{cases}$

(d) $P(2<X<5)=P(\frac{2-1}{2}<\frac{X-1}{2}<\frac{5-1}{2})=P(0.5<\frac{X-1}{2}<2)=\Phi(2)-\Phi(0.5)$

(e) ~~$P(X\ge 0)=1-P(X<0)=1-P(\frac{X-1}{2}<-\frac{1}{2})$~~

$P(X\ge 0)=P(\frac{X-1}{2}\ge -\frac{1}{2})=P(\frac{X-1}{2}\le\frac{1}{2})$

$=\Phi(0.5)$



或 $P(X\ge 0)=1-P(X<0)=1-P(\frac{X-1}{2}<-\frac{1}{2})=1-\Phi(-0.5)=1-[1-\Phi(0.5)]=\Phi(0.5)$

(f) $P(-1<X<0.5)=P(\frac{-1-1}{2}<\frac{X-1}{2}<\frac{0.5-1}{2})$

$=P(-1<\frac{X-1}{2}<-0.25)$

$=\Phi(-0.25)-\Phi(-1)=[1-\Phi(0.25)]-[1-\Phi(1)]=\Phi(1)-\Phi(0.25)$

(g) $P(|X|\le 2)=P(-2\le X\le 2)=P(\frac{-2-1}{2}\le\frac{X-1}{2}\le\frac{2-1}{2})=P(-1.5\le\frac{X-1}{2}\le 0.5)$

$=\Phi(0.5)-\Phi(-1.5)=\Phi(0.5)-[1-\Phi(1.5)]=\Phi(0.5)+\Phi(1.5)-1$

(h) $P(1\le -2X+3\le 8)=P(-2.5\le X\le 1)=P(\frac{-2.5-1}{2}\le\frac{X-1}{2}\le\frac{1-1}{2})=P(-1.75\le\frac{X-1}{2}\le 0)$

$=\Phi(0)-\Phi(-1.75)=\frac{1}{2}-[1-\Phi(1.75)]=\Phi(1.75)-0.5$

58 解: $X\sim N(5,2)$, 则 $f_X(x)=\frac{1}{\sqrt{2\pi}\cdot\sqrt{2}}e^{-\frac{(x-5)^2}{4}}$

(a) 法一: $Y=2X+4\Rightarrow X=\frac{Y-4}{2}$

$\therefore f_Y(y)=f_X(\frac{y-4}{2})\cdot\frac{1}{2}=\frac{1}{4\sqrt{2\pi}}e^{-\frac{(\frac{y-4}{2}-5)^2}{4}}=\frac{1}{4\sqrt{2\pi}}e^{-\frac{(y-14)^2}{16}}$

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法二：$E(Y)=E(2X+4)=2E(X)+4=14$

$Var(Y)=var(2X+4)=4var(X)=8$

$\therefore Y\sim N(14,8)$

则 $f_Y(y)=\frac{1}{\sqrt{2\pi}\cdot 2\sqrt{2}}e^{-\frac{(x-14)^2}{16}}=\frac{1}{4\sqrt{\pi}}e^{-\frac{(x-14)^2}{16}}$

(b) 见(a)中法二.

59. 解：$P(X<116)=P(\frac{X-\mu}{\sigma}<\frac{116-\mu}{\sigma})=\underbrace{\Phi(\frac{116-\mu}{\sigma})=0.2}_{\text{可知 } \frac{116-\mu}{\sigma}<0}=1-\Phi(\frac{\mu-116}{\sigma})$

注：有的同学没注意此处 $\frac{116-\mu}{\sigma}<0$.

$P(X<328)=P(\frac{X-\mu}{\sigma}<\frac{328-\mu}{\sigma})=\Phi(\frac{328-\mu}{\sigma})=0.9$

$\therefore \begin{cases}\Phi(\frac{\mu-116}{\sigma})=0.8\\ \Phi(\frac{328-\mu}{\sigma})=0.9\end{cases}$

查表得：$\begin{cases}\frac{\mu-116}{\sigma}=0.84\\ \frac{328-\mu}{\sigma}=1.28\end{cases}$ $\therefore \mu=200$, $\sigma^2=10000$

3.1 见及商榷：

解 (a)

X \ Y	1	2	3	4	5	6	7	8	9	10	11	12	$f_X(x)$
1	0	$\frac{1}{36}$	0	0	0	0	0	0	0	0	0	0	$\frac{1}{36}$
2			$\frac{1}{18}$	$\frac{1}{36}$									$\frac{3}{36}$
3				$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{36}$							$\frac{5}{36}$
4					$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{36}$					$\frac{7}{36}$
5						$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{36}$			$\frac{9}{36}$
6							$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{36}$	$\frac{11}{36}$
$f_Y(y)$	0	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	1

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(b)

X \ Y	1	2	3	4	5	6	$f_X(x)$
1	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{6}{36}$
2		$\frac{1}{18}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{6}{36}$
3			$\frac{1}{12}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{6}{36}$
4				$\frac{1}{9}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{6}{36}$
5					$\frac{5}{36}$	$\frac{1}{36}$	$\frac{6}{36}$
6						$\frac{1}{6}$	$\frac{6}{36}$
$f_Y(y)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$	1

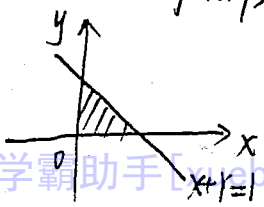
(c)

X \ Y	1	2	3	4	5	6	$f_X(x)$
1	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{11}{36}$
2		$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{9}{36}$
3			$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{7}{36}$
4				$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{5}{36}$
5					$\frac{1}{36}$	$\frac{1}{18}$	$\frac{3}{36}$
6						$\frac{1}{36}$	$\frac{1}{36}$
$f_Y(y)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$	1

3.4. 解: (a) $\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y)\,dx\,dy = c\int_0^{+\infty}\int_0^{+\infty} e^{-x-y}\,dx\,dy = c\int_0^{+\infty} e^{-x}dx \cdot \int_0^{+\infty} e^{-y}dy$

$= c\left(e^{-x}\Big|_0^{+\infty}\right)\left(e^{-y}\Big|_0^{+\infty}\right) = c = 1$

(b) $P(X+Y>1) = 1 - P(X+Y\le 1) = 1 - \int_0^1 dx \int_0^{1-x} e^{-x-y}\,dy = 2e^{-1}$



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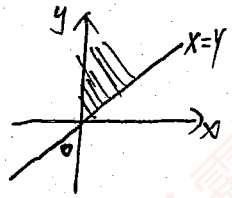
(c) $P(X<Y)=\int_0^{+\infty}dx\int_x^{+\infty}e^{-x-y}dy=\int_0^{+\infty}e^{-x}(-e^{-y}\big|_x^{+\infty})dx=\int_0^{+\infty}e^{-2x}dx=\frac{1}{2}$



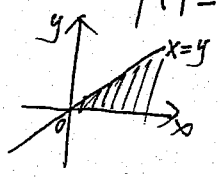
3.7. 解：(a) $\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}f(x,y)dxdy=1$

$=c\int_0^{+\infty}\int_0^{+\infty}e^{-ax}\cdot e^{-by}dxdy=c\int_0^{+\infty}e^{-ax}dx\int_0^{+\infty}e^{-by}dy=\frac{c}{ab}$

$\therefore C=ab$

(b) $P(Y\le X)=ab\int_0^{+\infty}e^{-ax}dx\int_0^{x}e^{-by}dy=a\int_0^{+\infty}e^{-ax}(1-e^{-bx})dx$

$=\frac{b}{a+b}$



(c) $x>0, y>0$ 时，$F(x,y)=ab\int_0^{x}e^{-au}du\int_0^{y}e^{-bv}dv$

$=e^{-au}\big|_0^{x}\cdot e^{-bv}\big|_0^{y}$

$=(1-e^{-ax})(1-e^{-by})$ 不少同学算错.

$\therefore F(x,y)=\begin{cases}(1-e^{-ax})(1-e^{-by}), & x>0, y>0\\ 0, & 其他\end{cases}$

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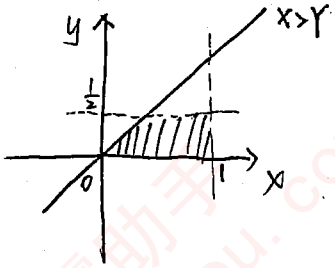
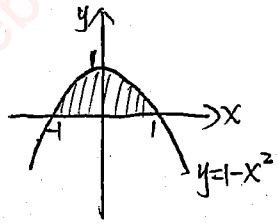
11. 解:



$\because X \perp\!\!\!\perp Y \quad \therefore f(x,y)=f_X(x)\cdot f_Y(y)$

$\therefore P(X>Y)=\int_0^{\frac{1}{2}} 8y\,dy \int_y^1 dx = \frac{2}{3}$

12. 解:



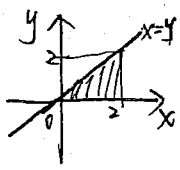
(a) $f_X(x)=\int_0^{1-x^2} \frac{15}{4}x^2\,dy = \frac{15}{4}x^2(1-x^2)$

$f_Y(y)=\int_{-\sqrt{1-y}}^{\sqrt{1-y}} \frac{15}{4}x^2\,dx = \frac{5}{2}(1-y)^{\frac{3}{2}}$

(b) $\because f(x,y)\neq f_X(x)\cdot f_Y(y)$

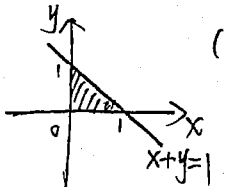
$\therefore$ X与Y不独立

14. 解: (a) $f(x,y)=f_X(x)\cdot f_Y(y)=\begin{cases}\frac{9}{64}x^2y^2, & 0\le x\le 2,\ 0\le y\le 2\\ 0, & \text{otherwise}\end{cases}$



(b) $P(X=Y)=0$ 注: 对同学得出 $P(X=Y)=\int_0^2 \frac{9}{64}x^4\,dx=\frac{9}{10}$

(c) $P(X>Y)=\int_0^2 dx\int_0^x \frac{9}{64}x^2y^2\,dy=\frac{1}{2}$



(d) $P(X+Y\le 1)=\int_0^1 dx\int_0^{1-x}\frac{9}{64}x^2y^2\,dy=\int_0^1 \frac{3}{64}x^2(1-x)^3\,dx=\frac{1}{1280}$

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21. 解: (a) $f_Y(y)=\int_0^1 \frac{12}{5}x(2-x-y)\,dx=\frac{8-6y}{5}$, $0<y<1$.

当$0<y<1$时, $f_{X|Y}(x,y)=\frac{f(x,y)}{f_Y(y)}=\begin{cases}\frac{6x(2-x-y)}{4-3y}, & 0<x<1\\ 0, & \text{otherwise}\end{cases}$

注: 都要写出范围是的和

(b) $P(X>\frac{1}{2}\mid Y=\frac{1}{3})=\int_{\frac{1}{2}}^1 f_{X|Y}(x|y=\frac{1}{3})\,dx=\int_{\frac{1}{2}}^1 \frac{6x(2-x-\frac{1}{3})}{4-1}\,dx=\frac{2}{3}$

23. 解:

(a) $f_Y(y)=\int_0^y 15x^2y\,dx=5y^4$, $0\le y\le 1$

$\because$ ~~当y<1时~~, $f_{X|Y}(x,y)=\frac{f(x,y)}{f_Y(y)}=\begin{cases}\frac{3x^2}{y^3}, & 0\le x\le y\le 1\\ 0 & \text{otherwise}\end{cases}$

(b) $f_X(x)=\int_x^1 15x^2y\,dy=\frac{15}{2}x^2(1-x^2)$, $0\le x\le 1$

注: 不同的不要x, y的范围

$f_{Y|X}(y|x)=\frac{f(x,y)}{f_X(x)}=\begin{cases}\frac{2y}{1-x^2}, & 0\le x\le y\le 1\\ 0 & \text{otherwise}\end{cases}$

(c) $\because f(x,y)\ne f_X(x)f_Y(y)$

$\therefore$ X与Y不独立.

24. 解: $f(x)=\begin{cases}1, & 0\le x\le 1\\ 0, & \text{otherwise}\end{cases}$ $\quad f_{Y|X}(y|x)=\begin{cases}\frac{1}{x}, & 0\le y\le x\\ 0, & \text{otherwise}\end{cases}$

(a) $f(x,y)=f_{Y|X}(y|x)f(x)=\begin{cases}\frac{1}{x}, & 0\le y\le x\le 1\\ 0, & \text{otherwise}\end{cases}$

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(b) $f_Y(y) = \int_y^1 \frac{1}{x}\,dx = -\ln y, \quad 0<y<1$

(c) $f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \begin{cases} -\frac{1}{x\ln y}, & 0<y\le x<1 \\ 0, & \text{otherwise} \end{cases}$

注：不少同学的错，这样做的：

(a) $f_X(x) = \begin{cases} 1, & 0<x<1 \\ 0, & \text{otherwise} \end{cases}$ $\quad f_Y(y) = \begin{cases} \frac{1}{x}, & 0\le y\le x \\ 0, & \text{otherwise} \end{cases}$

$f(x,y) = f_X(x) f_Y(y) = \begin{cases} \frac{1}{x}, & 0<y\le x<1 \\ 0, & \text{otherwise} \end{cases}$

(b) $f_Y(y) = \begin{cases} \frac{1}{x}, & 0\le y\le x \\ 0, & \text{otherwise} \end{cases}$

(c) $f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \begin{cases} 1, & y\le x<1 \\ 0, & \text{otherwise.} \end{cases}$

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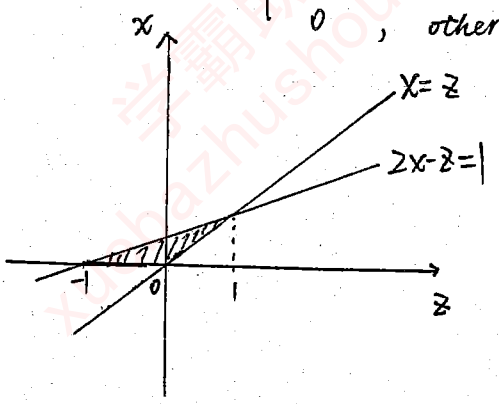
3.32. 解: $f_Z(z)=\int_{-\infty}^{+\infty} f(x, x-z)\,dx$

$$f(x, x-z)=\begin{cases} 6x, & x\ge 0,\ x-z\ge 0,\ 2x-z\le 1 \\ 0, & \text{otherwise} \end{cases}$$



注: 整体上的阴影部分

当 $-1\le z\le 0$ 时, $f_Z(z)=\int_0^{\frac{1+z}{2}} 6x\,dx=\frac{3}{4}(1+z)^2$

当 $0\le z\le 1$ 时, $f_Z(z)=\int_z^{\frac{1+z}{2}} 6x\,dx=\frac{3}{4}(1+z)^2-3z^2$

$$\therefore f_Z(z)=\begin{cases} \frac{3}{4}(1+z)^2-3z^2, & 0\le z\le 1 \\ \frac{3}{4}(1+z)^2, & -1\le z\le 0 \\ 0, & \text{otherwise} \end{cases}$$

当 $0\le z\le 1$ 时, $F_Z(z)=\int_{-\infty}^{z} f_Z(u)\,du=\int_{-1}^{0}\frac{3}{4}(1+z)^2\,dz+\int_0^z\left[\frac{3}{4}(1+u)^2-3u^2\right]du=\frac{(1+z)^3}{4}-z^3$

$\left(=\frac{3}{4}z^2+\frac{3}{4}z-\frac{3}{4}z^3+\right.$

当 $-1\le z<0$ 时, $F_Z(z)=\int_{-1}^{z}\frac{3}{4}(1+u)^2\,du=\frac{(1+z)^3}{4}$ $\left(=\frac{1}{4}z^3+\frac{3}{4}z^2+\frac{3}{4}z+\frac{3}{4}\right)$

$$\therefore F_Z(z)=\begin{cases} 1, & z\ge 1 \\ \frac{(1+z)^3}{4}-z^3, & 0\le z\le 1 \\ \frac{(1+z)^3}{4}, & -1\le z<0 \\ 0, & z<-1 \end{cases}$$

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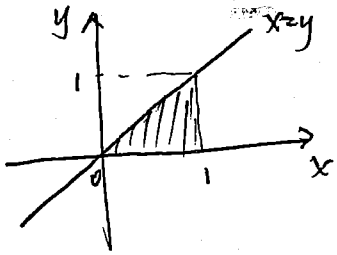
3.40

解: $f(x,y)=\begin{cases}\frac{1}{x}, & 0<y<x<1\\ 0, & \text{otherwise}\end{cases}$



(a) $f_Y(y)=\int_{-\infty}^{+\infty} f(x,y)dx=\int_y^1 \frac{1}{x}dx=-\ln y$, $0<y<1$

(b) $f_X(x)=\int_{-\infty}^{+\infty} f(x,y)dy=\int_0^x \frac{1}{x}dy=1$, $0<x<1$

(c) $E(X)=\int_{-\infty}^{+\infty} x f_X(x)dx=\int_0^1 x\,dx=\frac{1}{2}$ ~~$\frac{y^2}{2}$, $0<y<x$~~

(d) $E(Y)=\int_{-\infty}^{+\infty} y f_Y(y)dy=-\int_0^1 y\ln y\,dy=$ ~~$-\frac{x^2}{2}\ln x+\frac{x^2}{4}$, $y<x<1$~~

$=\frac{1}{4}$

注: 我同学把x, y的范围写错, 得到了我没画划掉部分之前的结果.

3.42.

解: $f(x)=\begin{cases}\frac{1}{b}, & x\in[0,b]\\ 0, & \text{otherwise}\end{cases}$

$E(X)=\frac{b}{2}$, $Var(X)=\frac{b^2}{12}$, $E(X^2)=Var(X)+(EX)^2=\frac{b^2}{3}$

(a) $Cov(X,X^2)=E(X^3)-E(X)\cdot E(X^2)$ 其中 $E(X^3)=\int_0^b \frac{x^3}{b}dx=\frac{b^3}{4}$

$=\frac{b^3}{4}-\frac{b}{2}\cdot\frac{b^2}{3}=\frac{b^3}{12}$

(b) $Var(X^2)=\int_0^b (x^2-\frac{b^2}{3})^2\frac{1}{b}dx=\frac{4b^2}{45}$

$\rho(X,X^2)=\frac{Cov(X,X^2)}{\sqrt{Var(X)}\sqrt{Var(X^2)}}=\frac{\frac{b^3}{12}}{\sqrt{\frac{b^2}{12}}\cdot\sqrt{\frac{4b^2}{45}}}=\frac{\sqrt{15}}{4}$

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44. $X \sim N(0,1)$, $Y \sim N(0,1)$

解：∵ X与Y独立 ∴ $cov(X,Y)=0$.

(a) $cov(3X+2Y, X+5Y+10)$

$= 3var(X) + \underbrace{15cov(X,Y) + 2cov(X,Y)}_{=0} + 10var(Y)$

$=13$

(b) $U = X+4Y \sim N(0,17)$

$P(X+4Y \geq 2) = P(U \geq 2) = P(\frac{U}{\sqrt{17}} \geq \frac{2}{\sqrt{17}}) = 1-\Phi(\frac{2}{\sqrt{17}})$

(c) $Z = X-Y$, $Z \sim N(0,2)$

$P((X-Y)^2>9) = P(Z^2>9) = P(Z>3) + P(Z<-3) = 2P(Z>3)$

$= 2P(\frac{Z}{\sqrt{2}} > \frac{3}{\sqrt{2}}) = 2[1-\Phi(\frac{3}{\sqrt{2}})]$

48. 解：$X \sim N(0,1)$, $Y \sim N(0,1)$ 且 X与Y独立.

则 $U = 0.6X + 0.8Y \sim N(0,1)$

(a) $corr(X,U) = \frac{cov(X,U)}{\sqrt{var(X)}\sqrt{var(U)}} = cov(X,U) = cov(X, 0.6X+0.8Y)$

$= 0.6\,var(X) = 0.6$

(b) 设 $U = aX+bY$，则 $U \sim N(0, a^2+b^2)$

$corr(X,U) = \frac{cov(X,U)}{\sqrt{var(X)}\sqrt{var(U)}} = \frac{E[aX^2+bXY]}{\sqrt{a^2+b^2}} = \frac{a}{\sqrt{a^2+b^2}} = \rho$

∴ 对给定的ρ，由公式 $\rho = \frac{a}{\sqrt{a^2+b^2}}$ 确定出 a, b，即可得到满足corr(X,U)的U.

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4.10.

Pf: $\bar{X} = \frac{S_n}{n}$, $E(\bar{X}) = p$, $Var(\bar{X}) = \frac{p(1-p)}{n}$

$$P(|\frac{S_n}{n} - p| \geq \varepsilon) = P(|\bar{X} - E(\bar{X})| \geq \varepsilon) \leq \frac{Var(\bar{X})}{\varepsilon^2} = \frac{p(1-p)}{n\varepsilon^2}$$

4.11 Pf: $X_i \sim U(0,1)$, $S_n = \sum_{i=1}^{n} X_i$

$$E(\frac{S_n}{n}) = \frac{\sum_{i=1}^{n} E(X_i)}{n} = \frac{1}{2}, \quad Var(\frac{S_n}{n}) = \frac{n\,Var(X_i)}{n^2} = \frac{1}{12n}$$

$$P(|\frac{S_n}{n} - \frac{1}{2}| \geq \varepsilon) \leq \frac{Var(\frac{S_n}{n})}{\varepsilon^2} = \frac{1}{12n\varepsilon^2}$$

4.12 $X \sim B(900, \frac{1}{2})$, $E(X) = 450$, $Var(X) = 225 = 15^2$

$$P(420 \leq X \leq 465) = P(\frac{420-450}{15} \leq \frac{X-450}{15} \leq \frac{465-450}{15})$$

$$= P(-2 \leq Z \leq 1)$$

$$= \Phi(1) - \Phi(-2) = \Phi(1) + \Phi(2) - 1 = 0.8185$$

4.16. (a) $k \sim B(1000, 0.6)$

$$P(550 \leq k \leq 650) = P(\frac{550-600}{\sqrt{240}} \leq \frac{k-600}{\sqrt{240}} \leq \frac{650-600}{\sqrt{240}})$$

$$= P(\frac{-50}{\sqrt{240}} \leq Z \leq \frac{50}{\sqrt{240}})$$

$$= 2\Phi(\frac{50}{\sqrt{240}}) - 1$$

$$= 0.999$$

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(b) $k \sim B(n, 0.6)$

$$P(0.59n \le k \le 0.61n) = P\left(\frac{0.59n-0.6n}{\sqrt{0.24n}} \le \frac{k-0.6n}{\sqrt{0.24n}} \le \frac{0.61n-0.6n}{\sqrt{0.24n}}\right)$$

$$= P\left(-\frac{0.01n}{\sqrt{0.24n}} \le Z \le \frac{0.01n}{\sqrt{0.24n}}\right)$$

$$= 2\Phi\left(\frac{0.01\sqrt{n}}{\sqrt{0.24}}\right) - 1 = 0.95$$

$$\Phi\left(\frac{0.01\sqrt{n}}{\sqrt{0.24}}\right) = 0.975$$

注：如果用常数取为 $\frac{0.01\sqrt{n}}{\sqrt{0.24}} \approx 0.02\sqrt{n}$

得 $2\Phi(0.02\sqrt{n}) - 1 = 0.95$

$\Rightarrow n = 9604$

$\Rightarrow \quad n = 9220$

5.3. $X \sim N(2, 4)$，则 $X_t = X\cos w_0 t \sim N(2\cos w_0 t,\ 4\cos^2 w_0 t)$

$$\therefore f_{X_t}(x) = \frac{1}{\sqrt{2\pi}\cdot 2\cos w_0 t} e^{-\frac{(x-2\cos(w_0 t))^2}{2\cdot 4\cos^2 w_0 t}} = \frac{1}{2\sqrt{2\pi}\cos w_0 t} e^{-\frac{(x-2\cos(w_0 t))^2}{8\cos^2 w_0 t}}$$

注：不妨简单得 $f(x) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{(x-2)^2}{8}} \cdot \cos w_0 t$ $\quad (\cos w_0 t \ne 0$

5.5. $A \sim EXP(3)$，$E[A] = \frac{1}{3}$，$var[A] = \frac{1}{9}$

$\theta \sim U(0, 2\pi)$，$f(x) = \begin{cases} \frac{1}{2\pi}, & 0 \le x \le 2\pi \\ 0, & \text{otherwise} \end{cases}$，$E[\theta] = \pi$，$var[\theta] = \frac{\pi^2}{3}$

$$\mu_X(t) = E[X_t] = E[A\cos(w_0 t + \theta)] = E[A]\cdot E[\cos(w_0 t+\theta)] = \frac{1}{3}\cdot\frac{1}{2\pi}\int_0^{2\pi}\cos(w_0 t+\theta)\,d\theta = 0$$

$$R_X(t_1, t_2) = E[A\cos(w_0 t_1+\theta)\cdot A\cos(w_0 t_2+\theta)]$$

$$= E[A^2]\cdot E[\cos(w_0 t_1+\theta)\cos(w_0 t_2+\theta)]$$

$$= [var(A) + E^2(A)]\cdot\frac{1}{2\pi}\int_0^{2\pi}\cos(w_0 t_1+\theta)\cos(w_0 t_2+\theta)\,d\theta$$

$$= \frac{2}{9}\cdot\frac{1}{2\pi}\int_0^{2\pi}\frac{1}{2}[\cos(w_0 t_1 + w_0 t_2 + 2\theta) + \cos(w_0 t_1 - w_0 t_2)]$$

$$= \frac{1}{9\cdot 2\pi}\left[\int_0^{2\pi}\cos(w_0 t_1 + w_0 t_2 + 2\theta)\,d\theta + \int_0^{2\pi}\cos(w_0 t_1 -\right.$$

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$$= \frac{1}{9\cdot 2\pi}\cdot 2\pi\cos(w_0(t_1-t_2)) = \frac{1}{9}\cos w_0(t_1-t_2)$$

5.6. (a) $\mu_X(t) = E[X_t] = E[At] = tE[A] = 0$

$\sigma_X^2(t) = var(X_t) = var(At) = t^2 var(A) = 4t^2$

$$f_{X_t}(x) = \frac{1}{\sqrt{2\pi}\cdot 2t} e^{-\frac{x^2}{8t^2}} = \frac{1}{2t\sqrt{2\pi}} e^{-\frac{x^2}{8t^2}}$$

(b) $\mu_X(t) = E[At] = tE[A] = 0$

$R_X(t_1,t_2) = E[At_1\cdot At_2] = t_1t_2E[A^2] = t_1t_2[var(A)+E^2(A)] = 4t_1t_2$

$C_X(s,t) = R_X(s,t) - \mu_X(s)\mu_X(t) = 4st$

$\sigma_X^2(t) = C_X(t,t) = 4t^2$

5.10.

$X_0 \sim Exp(1)$, $E(X_0)=1$, $var(X_0)=1$

$\theta \sim U(0,2\pi)$, $f(\theta) = \begin{cases} \frac{1}{2\pi}, & \theta\in(0,2\pi) \\ 0, & \text{otherwise} \end{cases}$, $E(\theta)=\pi$, $var(\theta)=\frac{\pi^2}{3}$

$\mu_X(t) = E[X_t] = E[X_0] + E[\cos(2\pi t+\theta)] = 1 + \frac{1}{2\pi}\int_0^{2\pi}\cos(2\pi t+\theta)d\theta = 1$

$R_X(s,t) = E[X_sX_t] = E[(X_0+\cos(2\pi s+\theta))(X_0+\cos(2\pi t+\theta))]$

$= E[X_0^2] + E[X_0]E[\cos(2\pi t+\theta)] + E[X_0]E[\cos(2\pi s+\theta)] + E[\cos(2\pi s+\theta)\cos(2\pi t+\theta)]$

$= Var(X_0) + E^2(X_0) + \underbrace{\frac{1}{2}\cos(2\pi(s-t))}_{\text{由上题(5.6)知}}$

$= 2 + \frac{1}{2}\cos 2\pi(s-t)$

很多同学没算得不对.

$\sigma_X^2(t) = R_X(t,t) - \mu_X^2(t) = 2+\frac{1}{2}-1 = \frac{3}{2}$

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1. $P=\begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 \end{pmatrix}$, $P(X_0=0)=1$, $P(X_0=1)=0$, $P(X_0=2)=0$.

解：(a) $P(X_0=0, X_1=1, X_2=2)=P(X_0=0)\cdot P(X_1=1|X_0=0)\,P(X_2=2|X_1=1)$

$$=1\cdot P_{01}\cdot P_{12}=\frac{1}{4}$$

(b) $P^{(3)}=P\cdot P\cdot P=\begin{pmatrix} \frac{1}{8} & \frac{3}{4} & \frac{1}{8} \\ \frac{3}{16} & \frac{5}{8} & \frac{3}{16} \\ \frac{1}{8} & \frac{3}{4} & \frac{1}{8} \end{pmatrix}$

$\therefore\ P(X_3=1)=\sum_{i=0}^{2}P(X_3=1|X_0=i)P(X_0=i)=P(X_3=1|X_0=0)P(X_0=0)$

$$=P_{01}^{(3)}\times 1=\frac{3}{4}$$

2. $S=\{1,2,3,4\}$, $P(X_0=1)=P(X_0=2)=P(X_0=3)=P(X_0=4)=\frac{1}{4}$, $P=\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{4} & \frac{3}{4} & 0 \\ \frac{1}{4} & 0 & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix}$

(a) $P^{(4)}=P^{(2)}\cdot P^{(2)}=\begin{bmatrix} \frac{1}{4} & \frac{3}{8} & \frac{3}{8} & 0 \\ \frac{3}{16} & \frac{1}{16} & \frac{9}{16} & \frac{3}{16} \\ \frac{3}{8} & \frac{1}{8} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & 0 & \frac{1}{4} \end{bmatrix}\begin{bmatrix} \frac{1}{4} & \frac{3}{8} & \frac{3}{8} & 0 \\ \frac{3}{16} & \frac{1}{16} & \frac{9}{16} & \frac{3}{16} \\ \frac{3}{8} & \frac{1}{8} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & 0 & \frac{1}{4} \end{bmatrix}=\begin{bmatrix} \frac{35}{128} & \frac{21}{128} & \frac{51}{128} & \frac{21}{128} \\ \frac{93}{256} & \frac{49}{256} & \frac{63}{256} & \frac{51}{256} \\ \frac{86}{256} & \frac{62}{256} & \frac{70}{256} & \frac{38}{256} \\ \frac{76}{256} & \frac{68}{256} & \frac{84}{256} & \frac{28}{256} \end{bmatrix}$

$P(X_4=1)=\frac{1}{4}\left(P_{11}^{(4)}+P_{21}^{(4)}+P_{31}^{(4)}+P_{41}^{(4)}\right)=\frac{325}{1024}$

$P(X_4=2)=\frac{1}{4}\left(P_{12}^{(4)}+P_{22}^{(4)}+P_{32}^{(4)}+P_{42}^{(4)}\right)=\frac{221}{1024}$

$P(X_4=3)=\frac{1}{4}\left(P_{13}^{(4)}+P_{23}^{(4)}+P_{33}^{(4)}+P_{43}^{(4)}\right)=\frac{319}{1024}$

$P(X_4=4)=\frac{1}{4}\left(P_{14}^{(4)}+P_{24}^{(4)}+P_{34}^{(4)}+P_{44}^{(4)}\right)=\frac{159}{1024}$

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$P(X_2=2, X_3=1, X_5=4) = P(X_2=2)\cdot P_{21}\cdot P_{14}^{(2)}$ 其中 $P_{14}^{(2)}=0$ $=0$

(c)

$$P^{(3)} = \begin{pmatrix} \frac{16}{64} & \frac{16}{64} & \frac{30}{64} & \frac{6}{64} \\ \frac{21}{64} & \frac{7}{64} & \frac{21}{64} & \frac{15}{64} \\ \frac{24}{64} & \frac{16}{64} & \frac{16}{64} & \frac{12}{64} \\ \frac{24}{64} & \frac{20}{64} & \frac{12}{64} & \frac{8}{64} \end{pmatrix}$$

$$P(X_3=1, X_5=2 \mid X_0=3) = P(X_3=1 \mid X_0=3)\cdot P(X_5=2 \mid X_3=1)$$

$$= P_{31}^{(3)}\cdot P_{12}^{(2)} = \frac{24}{64}\cdot\frac{3}{8} = \frac{9}{64}$$

3. (a) $P=\begin{pmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{pmatrix}$ (b) $P=\begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$ $S=\{1,2,3\}$

解 (a) $\because \forall x,y\in S$, 有 $P_{xy}(1)>0$ $\therefore$ 是遍历的.

$$\begin{cases} (\pi_0,\pi_1,\pi_2)=(\pi_0,\pi_1,\pi_2)P \\ \pi_0+\pi_1+\pi_2=1 \\ \pi_i\geq 0,\ i=0,1,2 \end{cases}$$ 得 $\pi=(\frac{16}{35}, \frac{13}{35}, \frac{8}{35})$

(b) $\because P^2=\begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$, $\forall x,y\in S$, 有 $P_{xy}(2)>0$ $\therefore$ 是遍历的.

$$\begin{cases} (\pi_0,\pi_1,\pi_2)=(\pi_0,\pi_1,\pi_2)P \\ \pi_0+\pi_1+\pi_2=1 \\ \pi_i\geq 0,\ i=0,1,2 \end{cases}$$ 得 $\pi=(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

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(b) $P(X_2=2, X_3=1, X_5=4) = P(X_2=2)\cdot P_{21}\cdot P_{14}^{(2)}$ 其中 $P_{14}^{(2)}=0$ $\underline{\underline{\quad 0 \quad}}$

(c) $$P^{(3)} = \begin{pmatrix} \frac{14}{64} & \frac{14}{64} & \frac{30}{64} & \frac{6}{64} \\ \frac{21}{64} & \frac{7}{64} & \frac{21}{64} & \frac{15}{64} \\ \frac{24}{64} & \frac{14}{64} & \frac{14}{64} & \frac{12}{64} \\ \frac{24}{64} & \frac{20}{64} & \frac{12}{64} & \frac{8}{64} \end{pmatrix}$$

$$P(X_3=1, X_5=2 \mid X_0=3) = P(X_3=1 \mid X_0=3)\cdot P(X_5=2 \mid X_3=1)$$
$$= P_{31}^{(3)}\cdot P_{12}^{(2)} = \frac{24}{64}\cdot\frac{3}{8} = \frac{9}{64}$$

3. (a) $P=\begin{pmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{pmatrix}$ (b) $P=\begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$ $S=\{1,2,3\}$

解 (a) $\because \forall x,y\in S$，有 $P_{xy}(1)>0$ $\therefore$ 是遍历的.

$$\begin{cases} (\pi_0,\pi_1,\pi_2)=(\pi_0,\pi_1,\pi_2)P \\ \pi_0+\pi_1+\pi_2=1 \\ \pi_i\ge 0,\ i=0,1,2 \end{cases}$$ 得 $\pi=\left(\frac{14}{35}, \frac{13}{35}, \frac{8}{35}\right)$

(b) $\because P^2=\begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$，$\forall x,y\in S$，有 ~~$P_{xy}$~~ $P_{xy}(2)>0$ $\therefore$ 是遍历的.

$$\begin{cases} (\pi_0,\pi_1,\pi_2)=(\pi_0,\pi_1,\pi_2)P \\ \pi_0+\pi_1+\pi_2=1 \\ \pi_i\ge 0,\ i=0,1,2 \end{cases}$$ 得 $\pi=\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

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